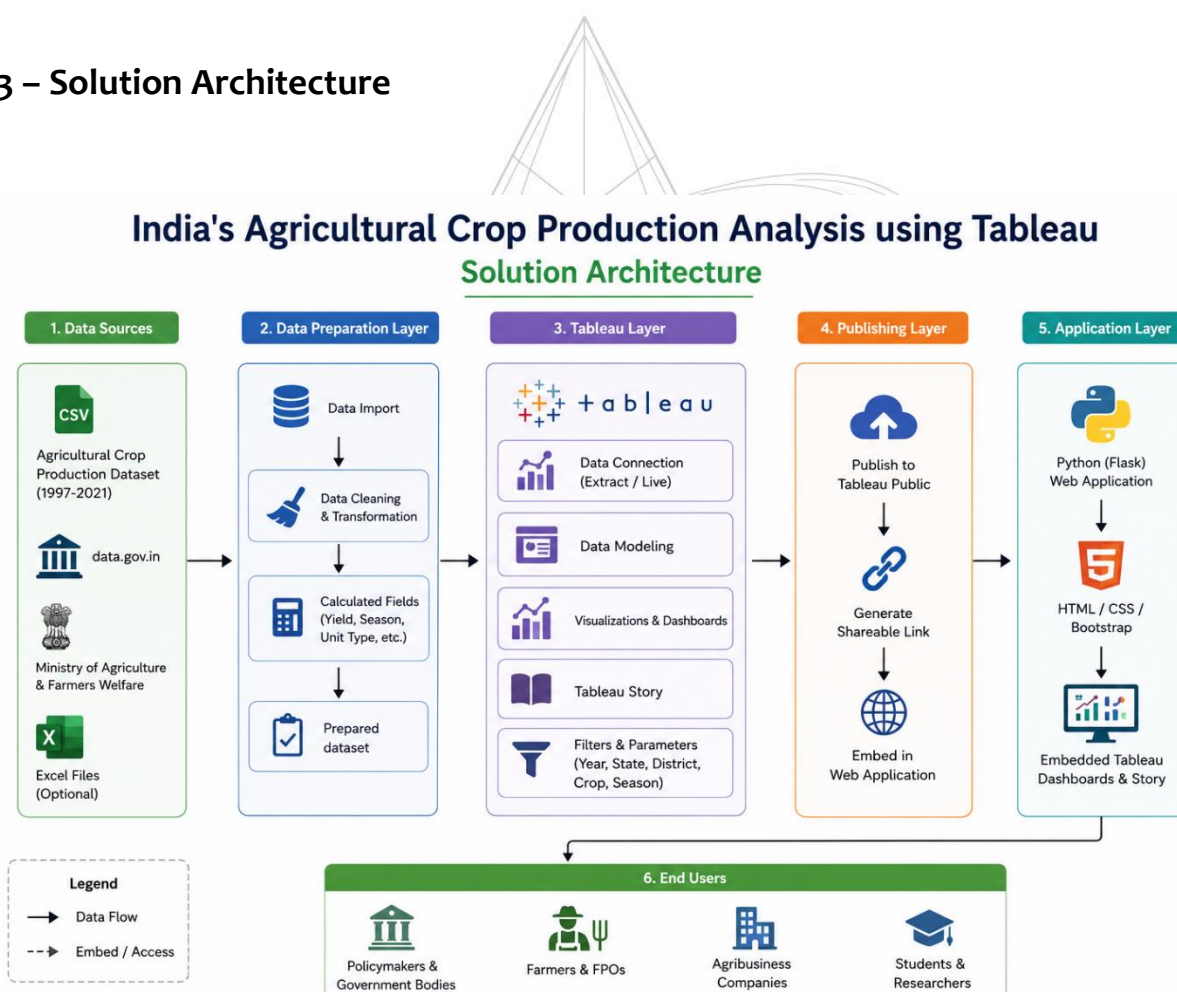


India's Agricultural Crop Production Analysis using Tableau

TEAM ID	SWTID-2026-7349
PROJECT TITLE	India's Agricultural Crop Production Analysis using Tableau
MAX MARKS	5 MARKS

4.3 – Solution Architecture



Data Sources

We need to get the crop production data from many places. These places include CSV files, the Ministry of Agriculture & Farmers Welfare portal, data.gov.in. Maybe some Excel files. We have to get the data for the years 1997 to 2021 for all the states, districts, crops and seasons.

Data Preparation Layer

Now we take the data and put it into a special place where we can clean it up and make it better. We remove any copies of the same data fix any missing information make sure the names of the crops and states are the same everywhere change the type of data when we need to and add some new fields that will help us analyze the data better. These new fields are things like how crop we get from a certain area what season it is and what kind of unit we are using.

Data Modeling

We take the cleaned-up data and organize it into a model. This model helps us see how all the different parts of the data are related to each other. We have things like the state, district, crop, season and year. We can see how they all connect to each other. We also make sure the data is connected in a way that makes it fast and easy to use.

Tableau Layer

Now we use the organized data to make pictures and graphs in Tableau. We make things like charts, graphs, maps and stories that people can use to explore the data. We add filters and parameters so people can look at the data for a year, state, district, crop or season.

Publishing Layer

We take the pictures and graphs we made and put them on Tableau Public so everyone can see them. We make links that people can use to share the data with others and we can even put it right into websites so it is easy to use.

Application Layer

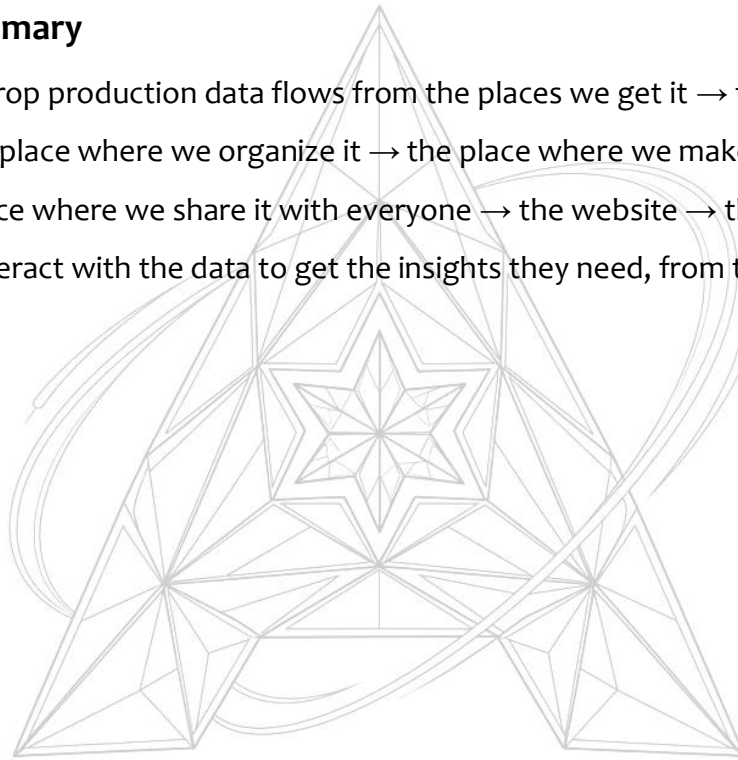
We use Python Flask to make a website that's easy to use on any device. We add the pictures and graphs from Tableau to the website so it all works together well.

End User Access

Now we let all the people who need to use the data have access to it. This includes people who make policies, farmers, companies that work with agriculture and students who want to learn from the data.

Data Flow Summary

The agricultural crop production data flows from the places we get it → the place where we clean it up → the place where we organize it → the place where we make pictures and graphs → the place where we share it with everyone → the website → the people who use it. They can all interact with the data to get the insights they need, from the agricultural crop production data.



ALPHA
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